

Benoît Pasquier

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Summary

I am a mathematically trained oceanographer and currently a Lecturer (Education Focused) at the Climate Change Research Centre (CCRC), UNSW. I specialise in numerical modelling of the marine biogeochemical cycles that govern the ocean's carbon, oxygen, and nutrient distributions, and how they respond to climate change. My research combines inverse modelling, linear and nonlinear dynamics, and Green-function diagnostics with rigorous data analysis and large climate-model archives. My work is consistently published in top-tier (Q1) peer-reviewed journals. I also develop open-source scientific software, including packages that lower the barrier to modelling ocean biogeochemistry for the wider community.

To study changes in ocean circulation, my current research extends transport-matrix methods (which turn ocean tracer transport into sparse linear algebra) to high-resolution ocean models. This work combines state-of-the-art nonlinear and large sparse linear solvers, automatic differentiation, and fluid-dynamics solvers distributed on multiple GPUs. My previous research, on the response of the ocean's carbon and oxygen cycles to a slower and warmer ocean, combined similar tools with parameter optimisation. Beyond novel computational pipelines, this work introduced new conceptual science relevant to pressing climate-change issues. This includes sequestration and ventilation timescales, of direct relevance to marine Carbon Dioxide Removal (mCDR), the biological pump, and ocean deoxygenation.

On teaching, I am redesigning CLIM1001/GENS0401, one of UNSW's largest courses, delivered fully online and asynchronously to about 1700 students in 2026 and still growing. I work with a small teaching team and lead the rebuild of its lectures, Moodle site, interactive widgets, and assessments in the age of AI. I have also lectured in CLIM3001/6001 (Climate Systems Science). Earlier, in the School of Mathematics and Statistics, I ran tutorials for MATH2089 (Numerical Methods and Statistics) with excellent student feedback, and provided lecture support and Weather-in-a-tank demonstrations for MATH3261/5285 (Geophysical Fluid Dynamics).

I plan to expand my research in two directions. First, by embedding modern statistical and machine-learning tools into computationally efficient global ocean biogeochemistry models. Second, by growing community-driven frameworks such as the Transport Matrix Intercomparison Project (TMIP) for diagnosing and comparing climate-model ocean circulation and biogeochemistry.

My background is broad and cross-disciplinary, spanning pure mathematics at École Polytechnique, mathematical finance at Paris-Dauphine/ENSAE, and applied mathematics/computational oceanography at UNSW, UCI, and USC. It equips me with a broad methodological toolkit, from stochastic calculus and uncertainty quantification to numerical optimisation and high-performance computing. It also allows me to teach across mathematics, scientific computing, and climate science.

Professional Experience

- Jun 26—Current **Lecturer (Education Focused)** CCRC, University of New South Wales, Sydney, Australia
Major rework of CLIM1001/GENS0401, one of UNSW's largest courses (online & asynchronous; about 1700 students over Trimesters 2 & 3 2026): Leading redesign of Moodle site, aligning content & assessments with learning outcomes, adding interactive widgets, reducing instructors' workload for marking, and building AI-resilient assessments.
Other activities and papers:
 - Giving in-person lectures for CLIM3001/6001.
 - Convening monthly CCRC AI discussion group.
 - Continuing transport-matrix research and TMIP coordination.
 - [Anthonisz et al., under review](#)
 - [Berg et al., *Paleoceanogr. Paleoclimatol.*, 2026](#)
- Sep 25—May 26 **Research Associate** CCRC, University of New South Wales, Sydney, Australia
High-resolution Southern Ocean diagnostics from transport matrices. Coordinating the emerging Transport Matrix Intercomparison Project (TMIP).
Supervisor: Prof. Laurie Menviel.
Papers:
 - [Bernish et al., *Front. Earth Sci.*, 2026](#)
 - [Wise et al., *GRL*, 2026](#)
- Sep 24—Feb 25 **Contract Researcher** CSIRO, Hobart, Australia
Built an offline transport-matrix framework that reduced multi-millennial sequestration simulations from years to hours of wall time, applied to the deep-ocean sequestration efficiency under climate-change scenarios.
Supervisor: Dr. Richard Matear.
Paper:
 - [Pasquier et al., *GRL*, 2025](#)
- Oct 21—Aug 24 **Research Associate** Maths & Stats, University of New South Wales, Sydney, Australia
Response of the ocean's carbon and oxygen cycles to climate change, with novel sequestration- and exposure-time diagnostics.
Supervisor: A/Prof. Mark Holzer.
Papers:
 - [Pasquier et al., *BG*, 2023](#)
 - [Pasquier et al., *BG*, 2024](#)
 - [Pasquier et al., *JGR Oceans*, 2024](#)
 - [Holzer et al., *GRL*, 2024](#)
 - [John et al., *GBC*, 2024](#)

- Nov 19–Oct 21 **Postdoctoral Researcher** University of Southern California, Los Angeles, CA, USA
Global marine trace-metal modelling (Nd, Ni); designed and released the GNOM optimised Nd-cycle model and contributed to AIBECS.jl.
Supervisor: A/Prof. Seth G. John.
Papers:
 - [Pasquier et al., GMD, 2022](#)
 - [John et al., Nat. Geosci., 2022](#)
 - [Holzer et al., GBC, 2021](#)
 - [Pasquier et al., JOSS, 2022](#)
- Sep 17–Sep 19 **Postdoctoral Research Scholar** University of California, Irvine, CA, USA
Designed and developed AIBECS.jl, an open-source Julia framework for global marine biogeochemistry modelling, used as a research and teaching tool across multiple groups.
Supervisors: Prof. François W. Primeau and Prof. J. Keith Moore.
Papers:
 - [Holzer et al., GBC, 2019](#)
 - [Meskhidze et al., Mar. Chem., 2019](#) (Iron at the Air–Sea Interface workshop)
 - [Britten et al., Front. Environ. Sci., 2021](#)
- Mar 17–Aug 17 **Casual Research Assistant** Maths & Stats, University of New South Wales, Sydney, Australia
PhD wrap-up.
Supervisor: A/Prof. Mark Holzer.
Papers:
 - [Pasquier and Holzer, BG, 2017](#) (P–Si–Fe inverse model)
 - [Pasquier and Holzer, BG, 2018](#) (iron regenerations; in preparation)
- May 11–Aug 12 **Proposal Engineer** Degrémont, Suez Environnement, Sydney, Australia
Tendering project management for design, construction, maintenance, and operational contracts. Business development, liaison with clients, advertising on company capabilities.
- Jul 08–Jun 09 **Forex Trader Assistant** Société Générale Investment Banking, Paris, France
MASEF Internship in foreign exchange market (Forex). Research and software development in automated arbitrage using real-time high-frequency data.
Supervisors: Prof. Bruno Bouchard and Dr. Nicolas Boitout.
- Apr 07–Jul 07 **Mathematics Research Intern** École Polytechnique, Palaiseau, France
École Polytechnique Speciality (Mathematics) Internship at the Laurent Schwartz Mathematics Centre (CMLS). Research review on the Witt ring of quadratic forms.
Supervisor: Prof. Jean Lannes.
- Sep 04–Feb 05 **IT Intern** Bioforce, Lyon, France
Development of an ACCESS database to improve communication and management of Bioforce, which provides training and career advice in aid programs and logistics.
- Jul 06–Jul 06 **Assembly Line Worker (Internship)** Mecaplast, Monaco
École Polytechnique Industrial Placement.

Education

- 2013–2017 **PhD in Applied Mathematics** University of New South Wales, Sydney, Australia
Thesis title: *The Ocean's Global Iron, Phosphorus, and Silicon Cycles: Inverse Modelling and Novel Diagnostics*.
Supervisor: A/Prof. Mark Holzer.
- Global Biogeochemical Cycles, Global Biological Pump
 - Ecosystem Modelling & Biogenic Transport Modelling
 - Green Functions Techniques (Path Densities, Flow Rates, Time Scales)
 - Nonlinear Systems, Parameter Optimisation/Inverse Modelling
 - Iron Control on the Global Biological Pump
 - Southern Ocean Nutrient Trapping
- 2010 **MSc in Environmental Science** University of New South Wales, Sydney, Australia
Study of the nature of environmental problems and the methodology of their evaluation and management.
- Geophysical Fluid Dynamics
 - Oceanography
 - Project Management, Environmental Risk Management
- 2007–2008 **MSc in Finance Mathematics** Paris Dauphine + ENSAE ParisTech, Paris, France
MASEF (Mathematics of Insurance, Economics and Finance), Finance Specialty.
Supervisor: Prof. Bruno Bouchard.
- Stochastic Calculus, Levy Processes with Jumps
 - Stochastic Differential Equations
 - Numerical Methods (Monte Carlo)
- 2004–2007 **MSc in Mathematics & Engineering** École Polytechnique, Palaiseau, France
Pure mathematics specialisation.
- Algebra, Arithmetic, Numerical Methods, Computer Science
 - Differential Topology, Relativity
 - Physics, Biology
- 2001–2004 **Preparatory Classes** Lycée Masséna, Nice, France
French Preparatory Classes, mathematics Specialty.
- Linear Algebra, Topology, Numerical Methods, Computer Science
 - Mechanics, Electromagnetism, Thermodynamics

Publications

I have contributed to 21 scientific articles, including 9 first-author works (marked with *); 20 published and 1 currently under review. With 376 citations, my h-index is 10, and my i10-index is 11 ([Google Scholar](#)).

- 21 A Warm Stagnant Ocean Leads to Deep-Ocean Deoxygenation without Widespread Anoxia
Benjamin W. Anthonisz, David K. Hutchinson, Katrin J. Meissner, Samar Khatiwala, **Benoît Pasquier**
EGUsphere (2026, under review) pp. 1–41. Copernicus GmbH | doi: [10.5194/egusphere-2026-3378](https://doi.org/10.5194/egusphere-2026-3378)
- 20 Simulated Ocean Oxygen Under Miocene Boundary Conditions
James R. Berg, David K. Hutchinson, Katrin J. Meissner, **Benoît Pasquier**, Mark Holzer, Alexandra Auderset
Paleoceanography and Paleoclimatology (2026) e2026PA005505 | doi: [10.1029/2026PA005505](https://doi.org/10.1029/2026PA005505)
- 19 Using a Cell Flux Model to Investigate Phytoplankton Growth and Macromolecular Allocation under Iron Limitation
Margaret Bernish, Meng Gao, Jongsun Kim, **Benoît Pasquier**, Keisuke Inomura
Frontiers in Earth Science (2026). Frontiers | doi: [10.3389/feart.2026.1839921](https://doi.org/10.3389/feart.2026.1839921)
- 18 Differing Controls on the Cycling of Neodymium in the Atlantic and the Pacific
Paige M. Wise, **Benoît Pasquier**, Seth G. John, Sophia K. V. Hines
Geophysical Research Letters (2026) e2025GL117334 | doi: [10.1029/2025GL117334](https://doi.org/10.1029/2025GL117334)
- 17* **The Sequestration Efficiency of the Deep Ocean**
Benoît Pasquier, Richard J. Matear, Matthew A. Chamberlain, Tilo Ziehn, David K. Hutchinson, François W. Primeau, Yi Liu, Ann Bardin
Geophysical Research Letters (2025) e2025GL116799 | doi: [10.1029/2025GL116799](https://doi.org/10.1029/2025GL116799)
- 16* **Deoxygenation and Its Drivers Analyzed in Steady State for Perpetually Slower and Warmer Oceans**
Benoît Pasquier, Mark Holzer, Matthew A. Chamberlain, Richard J. Matear, Nathaniel L. Bindoff
Journal of Geophysical Research: Oceans (2024) e2024JC021043 | doi: [10.1029/2024JC021043](https://doi.org/10.1029/2024JC021043)
- 15* **The Biological and Preformed Carbon Pumps in Perpetually Slower and Warmer Oceans**
Benoît Pasquier, Mark Holzer, Matthew A. Chamberlain
Biogeosciences (2024) pp. 3373–3400. Copernicus GmbH | doi: [10.5194/bg-21-3373-2024](https://doi.org/10.5194/bg-21-3373-2024)
- 14 Atmospheric pCO₂ Response to Stimulated Organic Carbon Export: Sensitivity Patterns and Timescales
Mark Holzer, Tim DeVries, **Benoît Pasquier**
Geophysical Research Letters (2024) e2024GL108462 | doi: [10.1029/2024GL108462](https://doi.org/10.1029/2024GL108462)
- 13 Biogeochemical Fluxes of Nickel in the Global Oceans Inferred From a Diagnostic Model
Seth G. John, Hengdi Liang, **Benoît Pasquier**, Mark Holzer, Sam Silva
Global Biogeochemical Cycles (2024) e2023GB008018 | doi: [10.1029/2023GB008018](https://doi.org/10.1029/2023GB008018)
- 12* **Optimal Parameters for the Ocean's Nutrient, Carbon, and Oxygen Cycles Compensate for Circulation Biases but Replumb the Biological Pump**
Benoît Pasquier, Mark Holzer, Matthew A. Chamberlain, Richard J. Matear, Nathaniel L. Bindoff, François W. Primeau
Biogeosciences (2023) pp. 2985–3009. Copernicus GmbH | doi: [10.5194/bg-20-2985-2023](https://doi.org/10.5194/bg-20-2985-2023)

- 11 The Biogeochemical Balance of Oceanic Nickel Cycling
Seth G. John, Rachel L. Kelly, Xiaopeng Bian, Feixue Fu, M. Isabel Smith, Nathan T. Lanning, Hengdi Liang, **Benoît Pasquier**, Emily A. Seelen, Mark Holzer, Laura Wasylenki, Tim M. Conway, Jessica N. Fitzsimmons, David A. Hutchins, Shun-Chung Yang
Nature Geoscience (2022) pp. 906–912. Nature Publishing Group | doi: [10.1038/s41561-022-01045-7](https://doi.org/10.1038/s41561-022-01045-7)
- 10* **GNOM v1.0: An Optimized Steady-State Model of the Modern Marine Neodymium Cycle**
Benoît Pasquier, Sophia K. V. Hines, Hengdi Liang, Yingzhe Wu, Steven L. Goldstein, Seth G. John
Geoscientific Model Development (2022) pp. 4625–4656. Copernicus GmbH | doi: [10.5194/gmd-15-4625-2022](https://doi.org/10.5194/gmd-15-4625-2022)
- 9* **AIBECS.JI: A Tool for Exploring Global Marine Biogeochemical Cycles.**
Benoît Pasquier, François W. Primeau, Seth G. John
Journal of Open Source Software (2022) p. 3814. The Open Journal | doi: [10.21105/joss.03814](https://doi.org/10.21105/joss.03814)
- 8 A New Metric of the Biological Carbon Pump: Number of Pump Passages and Its Control on Atmospheric pCO₂
Mark Holzer, Eun Young Kwon, **Benoît Pasquier**
Global Biogeochemical Cycles (2021) e2020GB006863 | doi: [10.1029/2020GB006863](https://doi.org/10.1029/2020GB006863)
- 7 Evaluating the Benefits of Bayesian Hierarchical Methods for Analyzing Heterogeneous Environmental Datasets: A Case Study of Marine Organic Carbon Fluxes
Gregory L. Britten, Yara Mohajerani, Louis Primeau, Murat Aydin, Catherine Garcia, Wei-Lei Wang, **Benoît Pasquier**, B. B. Cael, François W. Primeau
Frontiers in Environmental Science (2021) | doi: [10.3389/fenvs.2021.491636](https://doi.org/10.3389/fenvs.2021.491636)
- 6 Diatom Physiology Controls Silicic Acid Leakage in Response to Iron Fertilization
Mark Holzer, **Benoît Pasquier**, Timothy DeVries, Mark Brzezinski
Global Biogeochemical Cycles (2019) pp. 1631–1653 | doi: [10.1029/2019GB006460](https://doi.org/10.1029/2019GB006460)
- 5 Perspective on Identifying and Characterizing the Processes Controlling Iron Speciation and Residence Time at the Atmosphere-Ocean Interface
Nicholas Meskhidze, Christoph Völker, Hind A. Al-Abadleh, Katherine Barbeau, Matthieu Bressac, Clifton Buck, Randelle M. Bundy, Peter Croot, Yan Feng, Akinori Ito, Anne M. Johansen, William M. Landing, Jingqiu Mao, Stelios Myriokefalitakis, Daniel Ohnemus, **Benoît Pasquier**, Ying Ye
Marine Chemistry (2019) p. 103704 | doi: [10.1016/j.marchem.2019.103704](https://doi.org/10.1016/j.marchem.2019.103704)
- 4* **The Number of Past and Future Regenerations of Iron in the Ocean and Its Intrinsic Fertilization Efficiency**
Benoît Pasquier, Mark Holzer
Biogeosciences (2018) pp. 7177–7203. Copernicus GmbH | doi: [10.5194/bg-15-7177-2018](https://doi.org/10.5194/bg-15-7177-2018)
- 3* **Inverse-Model Estimates of the Ocean's Coupled Phosphorus, Silicon, and Iron Cycles**
Benoît Pasquier, Mark Holzer
Biogeosciences (2017) pp. 4125–4159. Copernicus GmbH | doi: [10.5194/bg-14-4125-2017](https://doi.org/10.5194/bg-14-4125-2017)
- 2 The Age of Iron and Iron Source Attribution in the Ocean
Mark Holzer, Marina Frants, **Benoît Pasquier**
Global Biogeochemical Cycles (2016) pp. 1454–1474 | doi: [10.1002/2016GB005418](https://doi.org/10.1002/2016GB005418)
- 1* **The Plumbing of the Global Biological Pump: Efficiency Control through Leaks, Pathways, and Time Scales**
Benoît Pasquier, Mark Holzer
Journal of Geophysical Research: Oceans (2016) pp. 6367–6388 | doi: [10.1002/2016JC011821](https://doi.org/10.1002/2016JC011821)

Talks and Posters

I have contributed to at least 36 conference or seminar talks, posters, and tutorials, including 22 as a first author (marked with *), two invited talks at the MIT EAPS department, and one invited tutorial at the Woods Hole Oceanographic Institution (WHOI).

- 36*** **Fully equilibrated ventilation tracers across ACCESS-OM2 experiments: an ongoing journey**
Benoît Pasquier, Laurie Menviel, Gregory Wagner, Simone Silvestri, Alexis Montoisson, Taimoor Sohail, Andrew Kiss, François Primeau
ACEAS Seminar Series, 2026, Online | [Slides](#)
- 35*** **Building and running global marine biogeochemistry models with AIBECS.jl: a hands-on tutorial**
Benoît Pasquier
Invited to present at *Transport Matrix Intercomparison Project (TMIP) Inaugural Workshop*, 2026, Woods Hole Oceanographic Institution, Woods Hole, MA, USA
- 34 Research opportunities and risks with AI: a focus on AI coding
Alex Sen Gupta, **Benoît Pasquier**, Neil Malan
UNSW CCRC Fun Seminar, 2026, Sydney, Australia | [Slides](#)
- 33 Simulated Ocean Oxygen under Miocene Boundary Conditions
James Berg, David K. Hutchinson, Katrin Meissner, **Benoît Pasquier**, Mark Holzer, Alexandra Auderset
EGU General Assembly 2026, 2026, Vienna, Austria | doi: [10.5194/egusphere-egu26-5952](https://doi.org/10.5194/egusphere-egu26-5952)
- 32 Resolving the Top-Down/Bottom-Up Debate: A Basin-Specific Nd Isotope Cycle
Paige M. Wise, **Benoît Pasquier**, Seth G. John, Sophia K. V. Hines
Ocean Sciences Meeting, 2026, Glasgow, Scotland
- 31*** **Diagnosing the Ocean's Sequestration Efficiency Using Transport Matrices Built from Climate-Model Archives**
Benoît Pasquier, Richard J. Matear, Matthew A. Chamberlain, Tilo Ziehn, David K. Hutchinson, François W. Primeau, Yi Liu, Ann Bardin
Ocean Sciences Meeting, 2026, Online | [Slides](#) | [Video](#)
- 30*** **Diagnosing the ocean's sequestration efficiency using ACCESS-ESM1.5 transport matrices**
Benoît Pasquier, Richard J. Matear, Matthew A. Chamberlain, Tilo Ziehn, David K. Hutchinson, François W. Primeau, Yi Liu, Ann Bardin
ACCESS Community Workshop, 2025, Melbourne, Australia | [Slides](#)
- 29*** **The sequestration efficiency of the deep ocean: Fast computations using transport matrices built from ACCESS-ESM1.5 archives**
Benoît Pasquier, Richard J. Matear, Matthew A. Chamberlain, Tilo Ziehn, David K. Hutchinson, François W. Primeau, Yi Liu, Ann Bardin
COSIMA Working Group Meeting, 2025, Online | [PDF](#) | [Slides](#)
- 28 What Sets Neodymium Distributions in the Ocean and How Does This Impact its Utility as a Paleoceanographic Proxy?
Sophia K. V. Hines, Paige M. Wise, **Benoît Pasquier**, Seth G. John
AGU Fall Meeting, 2024, Washington, DC, USA
- 27 Shallow vs Deep: Identifying the magnitude and locations of marine Nd sources with GNOM
Paige M. Wise, **Benoît Pasquier**, Seth G. John, Sophia K. V. Hines
Goldschmidt Conference, 2024, Chicago, IL, USA

- 26* **The Ocean's Carbon and Oxygen Cycles in Future Steady-State Climate Scenarios**
Benoît Pasquier, Mark Holzer, Matthew A. Chamberlain, Richard J. Matear, Nathaniel L. Bindoff, François W. Primeau
Ocean Sciences Meeting, 2024, New Orleans, Louisiana, USA | [PDF](#) | [Slides](#)
- 25* **Optimal parameters for the ocean's nutrient, carbon, and oxygen cycles compensate for circulation biases but replumb the biological pump**
Benoît Pasquier, Mark Holzer, Matthew A. Chamberlain, Richard J. Matear, Nathaniel L. Bindoff, François W. Primeau
AMOS National Conference, 2024, Canberra, Australia | [PDF](#) | [Slides](#)
- 24 Quantifying the Dominant Fluxes of the Modern Marine Neodymium Cycle through Model Optimization
 Paige M. Wise, Lowell Stott, Seth G. John, **Benoît Pasquier**
AGU Fall Meeting, 2023, San Francisco, USA
- 23 Global distribution of nickel sources and sinks from a diagnostic model
 Seth G. John, Hengdi Liang, **Benoît Pasquier**, Mark Holzer, Sam Silva
Goldschmidt Conference, 2023, Lyon, France | doi: [10.7185/gold2023.20635](#)
- 22* **PCO2: A simple biogeochemistry model embedded in a simple ocean circulation model in matrix form**
Benoît Pasquier
UNSW Ocean Research Carnival, 2023, UNSW, Sydney, Australia | [PDF](#) | [Slides](#)
- 21 The Biogeochemical Balance that Controls Oceanic Nickel Cycling in the Modern and Past Oceans
 Seth G. John, Rachel Kelly, Xiaopeng Bian, Shun-Chung Yang, Feixue Fu, Magdalene I. Smith, Nathan Lanning, Hengdi Liang, **Benoît Pasquier**, Emily Seelen, Mark Holzer, Tim M. Conway, Jessica N. Fitzsimmons, David Hutchins
Goldschmidt Conference, 2022, Honolulu, Hawaii, USA | doi: [10.46427/gold2022.12708](#)
- 20 Modeling Marine Ecosystems At Multiple Scales Using Julia
 Gaël Forget, **Benoît Pasquier**, Zhen Wu
JuliaCon, 2021, Online | [Video](#) | [Notebook](#)
- 19 Global Controls on the Distribution of Nickel in the Oceans
 Seth G. John, Shun-Chung Yang, Xiaopeng Bian, Rachel Kelly, David Hutchins, Feixue Fu, Hengdi Liang, **Benoît Pasquier**, Emily Seelen
Goldschmidt Conference, 2020, Online | doi: [10.46427/gold2020.1217](#)
- 18 Metals, Models, and Isotopes: Insights into the Biogeochemical Cycling of Trace Nutrients in the Ocean
 Seth G. John, Tim M. Conway, Tom Weber, Tim DeVries, Alessandro Tagliabue, Hengdi Liang, **Benoît Pasquier**
Goldschmidt Conference, 2020, Online | doi: [10.46427/gold2020.1216](#)
- 17 Julia users and tools for oceanography
 Gaël Forget, **Benoît Pasquier**, Alexander Barth, Milan Klöwer, Ali Ramadan, Gregory L. Wagner, Navid Constantinou
Ocean Sciences Meeting, 2020, San Diego Convention Center, San Diego, California, USA | [Video](#)

- 16* **AIBECS.jl: the ideal tool for marine biogeochemistry modelling**
Benoît Pasquier, François Primeau
Ocean Sciences Meeting, 2020, San Diego Convention Center, San Diego, California, USA | [PDF](#)
- 15* **F-1 algorithm: Efficient differentiation through large steady-state problems**
Benoît Pasquier, François Primeau
Applied Maths Seminar, 2019, School of Mathematics and Statistics, UNSW, Australia | [Notebook](#)
- 14* **Introducing AIBECS.jl, a Julia package for creating global marine biogeochemistry models**
Benoît Pasquier, François Primeau, J. Keith Moore
CCRC Seminars, 2019, Climate Change Research Centre (CCRC), UNSW, Australia | [Notebook](#)
- 13 Southern Ocean silicic-acid leakage: Sensitivity to diatom physiology and its Si isotope signature analyzed in an inverse model
Mark Holzer, **Benoît Pasquier**, Tim DeVries, Mark Brzezinski
AMOS National Conference, 2019, Darwin, Australia
- 12* **The number of past and future regenerations of iron in the ocean and its intrinsic fertilization efficiency**
Benoît Pasquier, Mark Holzer
Invited to present at *Michael Follows Group Meeting*, 2019, MIT, USA | [PDF](#)
- 11* **Developing a new, open-source, user-friendly, fast, modular, global marine biogeochemistry model (in Julia)**
Benoît Pasquier
Invited to present at *Sack-lunch seminar*, 2019, MIT, USA | [PDF](#)
- 10* **Offline parameter optimization for global marine biogeochemical models**
Benoît Pasquier
François Primeau Group Meeting, 2018, University of California, Irvine, USA | [PDF](#)
- 9* **Inverse-model estimates of the ocean's coupled phosphorus, silicon, and iron cycles.**
Benoît Pasquier, Mark Holzer
Ocean Sciences Meeting, 2018, Portland, Oregon, USA | [PDF](#)
- 8* **The efficiency of different iron sources in supporting the ocean's global biological pump**
Benoît Pasquier, Mark Holzer
Half-baked seminar, Department of Earth System Science, 2017, University of California, Irvine, USA | [PDF](#)
- 7* **Response of the biological pump to perturbations in the iron supply: Global teleconnections diagnosed using an inverse model of the coupled phosphorus-silicon-iron nutrient cycles**
Benoît Pasquier, Mark Holzer
AMOS National Conference, 2017, Canberra, Australia | [PDF](#)
- 6* **Exploring iron control on global productivity: "FePSi", an inverse model of the ocean's coupled phosphate, silicon and iron cycles**
Benoît Pasquier, Mark Holzer
Postgrad Conference, 2016, Sydney, Australia | [PDF](#)
- 5 Iron Source Attribution and the Age of Dissolved Iron in the Ocean
Mark Holzer, **Benoît Pasquier**, Marina Frants
Ocean Sciences Meeting, 2016, New Orleans, Louisiana, USA

4* Iron control on global productivity: an efficient inverse model of the ocean's coupled phosphate, silicon, and iron cycles

Benoît Pasquier, Mark Holzer

Ocean Sciences Meeting, 2016, New Orleans, Louisiana, USA | [PDF](#)

3* The plumbing of the global biological pump

Benoît Pasquier, Mark Holzer

AMOS National Conference, 2015, Brisbane, Australia | [PDF](#)

2* An efficient inverse model of the ocean's coupled nutrient cycles

Benoît Pasquier, Mark Holzer

Postgrad Conference, 2015, Sydney, Australia | [PDF](#)

1* Plumbing of the biological pump

Benoît Pasquier, Mark Holzer

Postgrad Conference, 2014, Sydney, Australia | [PDF](#)

Teaching

- 2026–present **Instructor & Lecturer** *CLIM1001/GENS0401*, UNSW, Sydney, Australia
 Redesigning one of UNSW’s largest courses (about 1700 students in 2026, fully online and asynchronous). Rebuilt lectures and Moodle site, authored interactive web widgets, redesigned online quizzes and assessment for resilience to generative AI, and marked student work. Informal lead of a three-person teaching team.
- 2026 **Guest Lecturer** *CLIM3001/6001 (Climate Systems Science)*, UNSW, Sydney, Australia
 Two two-hour in-person lectures for undergraduate and postgraduate students: ocean biogeochemistry, and ocean and ice modelling.
- 2026 **Workshop Tutorials** TMIP inaugural workshop, WHOI, Woods Hole, MA, USA
 Two hands-on tutorials on AIBECS.jl and transport-matrix tooling for an international audience of ocean modellers; excellent participant feedback.
- 2021 **Live-demo Tutorial** JuliaCon 2021, online
Modelling Marine Ecosystems at Multiple Scales using Julia workshop. AIBECS.jl tutorial for an international research-software audience (organised by Dr. Gaël Forget, MIT).
- 2020 **Workshop Tutorial** Ocean Sciences Meeting, San Diego, CA, USA
Julia Users and Tools for Oceanography workshop. AIBECS.jl demonstration for the oceanographic community (organised by Dr. Gaël Forget, MIT).
- 2018–2019 **Substitute Lecturer & Notebook Author** UCI, Irvine, CA, USA
 Substituted lectures in Physical Oceanography (for Prof. Adam Martiny) and Biogeochemistry (for Prof. François Primeau) at postgraduate/PhD level.
 Designed AIBECS-based Jupyter notebooks for the Marine Biogeochemistry course (Prof. J. Keith Moore & Prof. François Primeau), introducing students to scientific programming and numerical exploration of ocean tracers.
- 2019–present **Tutorials & One-on-one Training** USC and UNSW
 Recorded short-video tutorials, live demonstrations, and one-on-one training in Julia, MATLAB, R, and biogeochemical modelling for postgraduate and PhD students at MTEL (USC) and the School of Mathematics and Statistics (UNSW). Sustained through the COVID-19 period as videoconference and recorded video lectures.
- Jun 16–Dec 16 **Mathematics Tutor** UNSW, Sydney, Australia
MATH2089 (Numerical Methods and Statistics; 2nd year). Computer-laboratory and theory tutorials, exam marking and supervision; excellent myExperience student feedback. Supervisor: Dr. Shev MacNamara.
- 2014–2017 **Teaching Assistance, Geophysical Fluid Dynamics** UNSW, Sydney, Australia
MATH3261/5285 (Geophysical Fluid Dynamics). Informal lecture support, setup and operation of the School’s Weather-in-a-tank demonstrations, and exam marking. Co-ran a Weather-in-a-tank session for the Year-10 Work Experience Day for students from across Sydney and regional NSW.
 Supervisor: A/Prof. Mark Holzer.
- ongoing **Open educational resources** AIBECS.jl, GNOM, and TMIP notebooks
 Author and maintainer of tutorials, documentation, and example notebooks for AIBECS.jl, GNOM, and the TMIP-code/notebooks repository, used as teaching material in graduate courses and adopted by independent researchers.

Mentoring, supervision, and technical help (informal)

- 2026—present **Informal team lead & staff mentor** CCRC, UNSW, Sydney, Australia
Coordinating a three-person team of new Education Focused hires on the redevelopment of CLIM1001/GENS0401: dividing the work, reviewing material, and onboarding colleagues to the course's tooling and workflows.
- 2025—present **Informal mentoring** CCRC, UNSW, Sydney, Australia
Informal mentoring of graduate students James Berg and Benjamin Anthonisz on coding workflows and ocean-modelling diagnostics. Resulted in two publications ([Berg et al., *Paleoceanogr. Paleoclimatol.*, 2026](#); [Anthonisz et al., *under review*](#))
- 2020—present **Co-mentor** MTEL, USC
Sustained mentoring of PhD student Paige Wise on the GNOM neodymium-cycle model: Code, methods, manuscript writing, and general career guidance. Resulted in co-authored presentations and publication ([Wise et al., *GRL*, 2026](#)).
Supervisor of record: A/Prof. Seth G. John.
- 2019—2021 **Tips, demos, and informal lectures** MTEL, USC
Coding tips, hands-on demos, and short presentations on Julia, MATLAB, and Jupyter notebooks to PhD students Hengdi Liang, Rachel Kelly, Phil Kongo, and Xiaopeng Bian.
- 2017—2019 **Coding tips and tricks** UCI, Irvine, CA, USA
Informal one-on-one help on Julia, MATLAB, git, and general coding to PhD students Faizan Haque, Gregory Britten, Qian Huang, and Christian Lewis (Department of Earth System Science).
- 2013—2017 **Peer help (during PhD)** School of Mathematics and Statistics, UNSW, Sydney, Australia
Technical mentoring of MSc student Yu-Heng (Ian) Ting (with A/Prof. M. Holzer); peer-level MATLAB help to fellow PhD student Carlos Rocha (with Prof. M. Roughan).

Open-source scientific software contributions

Owner	ClimateWidgets	https://github.com/briochemc/ClimateWidgets
	Interactive climate widgets built with Observable Framework (made for CLIM1001).	
Owner	academic-llm-guide	https://github.com/briochemc/academic-llm-guide
	A short, dated guide to generative AI tools for academic research and teaching.	
Owner	ACCESS-OM2_x_Oceananigans	https://github.com/TMIP-code/ACCESS-OM2_x_Oceananigans
	Work-in-progress code for the fastest pipeline for spinning-up tracers offline using Newton–Krylov solvers and running on multiple GPUs.	
Owner	OceanTransportMatrixBuilder.jl	https://github.com/TMIP-code/OceanTransportMatrixBuilder.jl
	A Julia package to build ocean transport matrices from CMIP model output.	
Owner	AIBECS.jl	https://github.com/JuliaOcean/AIBECS.jl
	The ideal tool for exploring global marine biogeochemical cycles.	
Owner	GNOM	https://github.com/MTEL-USC/GNOM
	An optimized steady-state model of the modern global marine neodymium cycle.	
Owner	F1Method.jl	https://github.com/briochemc/F1Method.jl
	Efficient quasi-auto-differentiation of an objective function defined implicitly by the solution of a steady-state problem.	
Owner	Inpaintings.jl	https://github.com/briochemc/Inpaintings.jl
	Julia version of MATLAB's inpaint_nans.	
Owner	WorldOceanAtlasTools.jl	https://github.com/briochemc/WorldOceanAtlasTools.jl
	Downloading and using data from the World Ocean Atlas (WOA) database.	
Owner	OceanographyCruises.jl	https://github.com/briochemc/OceanographyCruises.jl
	An interface for dealing with oceanographic cruises data.	
Owner	OceanGrids.jl	https://github.com/briochemc/OceanGrids.jl
	Standard format of grids for AIBECS.	
Owner	OceanBasins.jl	https://github.com/briochemc/OceanBasins.jl
	Programmatically determine which ocean basin a (lat,lon) coordinate is in.	
Owner	GEOTRACES.jl	https://github.com/briochemc/GEOTRACES.jl
	A package for reading and using GEOTRACES data in Julia.	
Owner	DualMatrixTools.jl	https://github.com/briochemc/DualMatrixTools.jl
	Efficiently solve dual-valued linear systems.	
Owner	HyperDualMatrixTools.jl	https://github.com/briochemc/HyperDualMatrixTools.jl
	Efficiently solve hyperdual-valued linear systems.	
Owner	BlockDiagonalFactors.jl	https://github.com/briochemc/BlockDiagonalFactors.jl
	Efficiently solve linear block-diagonal systems with repeated blocks.	
Owner	Earth2014.jl	https://github.com/briochemc/Earth2014.jl
	Download topographic data for the globe.	
Owner	ACCESS-TMIP	https://github.com/TMIP-code/ACCESS-TMIP
	Comparing diagnostics of water transport across ACCESS runs.	

Owner	TMIP-code/notebooks A place to explore and share TMIP-related Jupyter and Pluto notebooks.	https://github.com/TMIP-code/notebooks
Collaborator	UnitfulRecipes.jl Plotting data with units seamlessly in Julia.	https://github.com/jw3126/UnitfulRecipes.jl
Collaborator	HyperDualNumbers.jl Julia implementation of HyperDualNumbers.	https://github.com/JuliaDiff/HyperDualNumbers.jl
Contributor	Oceananigans.jl Julia software for fast, friendly, flexible, ocean-flavored fluid dynamics on CPUs and GPUs.	https://github.com/CliMA/Oceananigans.jl
Contributor	latexdiff Compares two latex files and marks up significant differences between them.	https://github.com/ftilmann/latexdiff
Contributor	Format.jl A Julia package to provide C and Python-like formatting support.	https://github.com/JuliaString/Format.jl
Contributor	cmap Perceptually uniform colormaps for MATLAB, compiled from multiple sources.	https://github.com/tsipkens/cmap
Contributor	Pardiso.jl Calling the PARDISO library from Julia.	https://github.com/JuliaSparse/Pardiso.jl
Contributor	YAXArrays.jl Yet Another XArray-like Julia package.	https://github.com/JuliaDataCubes/YAXArrays.jl
Contributor	Makie.jl Interactive data visualisations and plotting in Julia.	https://github.com/MakieOrg/Makie.jl
Contributor	offsetaxis-pkg OFFSETAXIS Add an x- or y-axis offset from the plotted axis area (MATLAB).	https://github.com/kakearney/offsetaxis-pkg
Contributor	GeoStats.jl Comprehensive framework for geostatistics (or spatial statistics).	https://github.com/JuliaEarth/GeoStats.jl
Contributor	AndersonAcceleration Anderson Acceleration for fixed point problems and spin-up of periodically-forced ocean models.	https://github.com/briochemc/AndersonAcceleration
Contributor	Plots.jl Powerful convenience for Julia visualisations and data analysis.	https://github.com/JuliaPlots/Plots.jl
Contributor	Unitful.jl Julia package for physical units.	https://github.com/PainterQubits/Unitful.jl
Contributor	UnitfulMoles.jl A set of predefined conventional elemental mol units.	https://github.com/briochemc/UnitfulMoles.jl
Contributor	Distributions.jl A Julia package for probability distributions and associated functions.	https://github.com/JuliaStats/Distributions.jl
Contributor	DiffEqBase.jl DiffEqBase.jl is a component package in the DiffEq ecosystem.	https://github.com/SciML/DiffEqBase.jl
Contributor	SciMLBase.jl The Base interface of the SciML ecosystem.	https://github.com/SciML/SciMLBase.jl

Contributor	OceanGreensFunctionMethods.jl	https://github.com/ggebbie/OceanGreensFunctionMethods.jl
	Green's Function Methods for Tracer Timescales and Pathways in Ocean Models.	
Contributor	DiffEqOperators.jl	https://github.com/SciML/DiffEqOperators.jl
	Linear operators for discretizations of differential equations and scientific machine learning (SciML).	
Contributor	Interpolations.jl	https://github.com/JuliaMath/Interpolations.jl
	Fast, continuous interpolation of discrete datasets in Julia.	
Contributor	RecipesBase.jl	https://github.com/JuliaPlots/RecipesBase.jl
	Base package for defining transformation recipes on user types for Plots.jl	
Contributor	CMAP.jl	https://github.com/simonscmap/CMAP.jl
	Simons CMAP Julia client.	
Contributor	payu	https://github.com/payu-org/payu
	A workflow management tool for numerical models on the NCI computing systems.	
Contributor	InverseDistanceWeighting.jl	https://github.com/juliohm/InverseDistanceWeighting.jl
	Inverse distance estimation solver for the GeoStats.jl framework.	
Contributor	NonlinearSolve.jl	https://github.com/SciML/NonlinearSolve.jl
	High-performance and differentiation-enabled nonlinear solvers (Newton methods), bracketed rootfinding (bisection, Falsi), with sparsity and Newton-Krylov support.	

Professional Service and Outreach

Peer review

ongoing **Reviewer** Selected peer-reviewed journals
Reviewer for journals including *Global Biogeochemical Cycles*, *Biogeosciences*, *Geoscientific Model Development*, *Journal of Geophysical Research: Oceans*, *Earth and Planetary Science Letters*, and *Nature Communications Earth & Environment*. Recognised as an Outstanding Reviewer by the AGU GBC editors in 2021.

Community engagement and organisation

2025–present **Co-initiator** Transport Matrix Intercomparison Project (TMIP)
Helping coordinate an international community effort to diagnose and compare ocean transport across climate models. Inaugural workshop held at WHOI in May 2026, where I led two hands-on tutorials.

2026–present **Initiator & Convenor** CCRC AI discussion group, UNSW, Sydney, Australia
Started and running a monthly centre-wide discussion group on the use of AI tools in research and teaching. Well attended and ongoing.

2017–2019 **Co-convenor & Co-chair** Department of Earth System Science, UC Irvine, CA, USA
Ran the weekly *Half-Baked Seminar* series for cross-disciplinary discussions among postgraduates, postdocs, and faculty with Christian Lewis.

2018–present **Open-source community helpdesk** Julia Slack, Discourse forum, Stack Overflow
Sustained activity answering scientific-computing, numerical-methods, and oceanographic-modelling questions on community helpdesks and forums.

ongoing **Active member** AGU, AMOS, COSIMA, OCB
Regular attendance and participation in scientific meetings and working groups (American Geophysical Union, Australian Meteorological and Oceanographic Society, Consortium for Ocean–Sea Ice Modelling in Australia, Ocean Carbon and Biogeochemistry).

Outreach

2024 **Climate-change presentation** Randwick Public School, Sydney, Australia
Outreach presentation introducing climate-change concepts to primary-school students.

ongoing **Open scientific software & reproducible research** AIBECS.jl, GNOM, OceanTransportMatrixBuilder.jl, etc.
Free, openly licensed, and documented scientific software lowering the barrier to studying global ocean biogeochemistry (see dedicated section).

Honors and Awards

2021	Outstanding Reviewer eos.org/agu-news/in-appreciation-of-agus-outstanding-reviewers-of-2021	AGU <i>Global Biogeochemical Cycles</i> editors
2015	Scholarship	Cuomo Foundation, Monaco
2014	Scholarship	Frères Louis et Max Principale Foundation, Monaco
2014--2016	Scholarship Higher studies scholarship	Monaco Government, Monaco
2013	Scholarship H.S.H. The Prince Albert II Exceptional Scholarship	Monaco Government, Monaco
2013--2016	Scholarship	Monaco Scientific Centre, Monaco
2013--2016	Tuition Fee Scholarship	Graduate Research School, UNSW, Sydney, Australia
2004--2008	Scholarship Higher studies scholarship	Monaco Government, Monaco

Skills

Research expertise

Marine biogeochemistry & global carbon cycle
Ocean transport matrices & Green-function diagnostics
Inverse modelling, parameter optimisation & UQ
Large sparse linear & nonlinear solvers
Automatic differentiation
Ocean deoxygenation & sequestration timescales
Earth System Model / CMIP analysis
Open scientific software & reproducible research

Teaching technologies

Moodle (course build & administration)
Online quiz banks & AI-resilient assessment
AI-assisted interactive web widgets
Jupyter & Pluto notebooks, live coding
Recorded video lectures & screencasts
Online / asynchronous delivery at scale
Git & GitHub-based course workflows

Programming

Julia / MATLAB	Expert
Python / LaTeX	Advanced
Bash / HPC scripting	Advanced
Git / GitHub workflows	Advanced
AI tools / Agentic coding	Advanced
NCO / CDO / NCL	Competent
Mathematica / Maple	Competent
HTML	Competent

Models & tools

ACCESS-ESM1.5, ACCESS-CM2, ACCESS-OM2
CESM1.2 (limited)
Transport Matrices (OCIM, TMIP)
Makie.jl, Plots.jl, SciML stack
NCI Gadi, Katana, university HPC clusters
xarray / dask / NumPy / SciPy

Professional References

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